CPU Scheduling Algorithms Analysis Report

Target Algorithms: Round Robin (RR), SRTF, SJF
Date: May 9, 2026
Project Scope: Comparative Study on Fairness vs. Efficiency

Mathematical Formulas

To ensure accuracy, the following standard Operating Systems formulas were used to calculate the performance metrics:

- **Completion Time (CT):** The exact time the process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
  - `TAT = Completion Time (CT) - Arrival Time (AT)`
- **Waiting Time (WT):** Total time a process spends waiting in the ready queue.
  - `WT = Turnaround Time (TAT) - Burst Time (BT)`
- **Response Time (RT):** Time from arrival until the first time the CPU is allocated.
  - `RT = First Start Time - Arrival Time (AT)`

Question 1

1. Round Robin (RR) Analysis (Quantum = 2)

The RR algorithm focuses on time-sharing fairness by giving each process a small fixed time slice (Quantum).

Metrics Table

Process	AT	BT	CT	TAT	WT	RT
P1	0	8	22	22	14	0
P2	1	4	12	11	7	1
P3	2	9	27	25	16	2
P4	3	6	24	21	15	5
Average				**19.75**	**13**	**2.00**

Round Robin Ready Queue (Execution Flow)

The ready queue manages process rotation as follows: `[P1] -> [P2, P1] -> [P3, P2, P1] -> [P4, P3, P2, P1] -> [P1, P4, P3, P2] -> ...`

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2. SRTF (Preemptive SJF) Analysis

SRTF selects the process with the shortest remaining execution time at any given point.

Metrics Table

Process	AT	BT	CT	TAT	WT	RT
P1	0	8	18	18	10	0
P2	1	4	5	4	0	0
P3	2	9	27	25	16	16
P4	3	6	11	8	2	2
Average				**13.75**	**7.00**	**4.50**

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3. SJF (Non-Preemptive) Analysis

In Non-Preemptive SJF, the CPU is held by a process until it completes, selecting the shortest available job only after the current one finishes.

Metrics Table

Process	AT	BT	CT	TAT	WT	RT
P1	0	8	8	8	0	0
P2	1	4	12	11	7	7
P3	2	9	27	25	16	16
P4	3	6	18	15	9	9
Average				**14.75**	**8.00**	**8.00**

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4. Comparison Summary

Metric	Round Robin (Q=2)	SRTF	SJF (Non-Preemptive)	Winner
Avg Waiting Time	13	**7.00**	8.00	**SRTF**
Avg Turnaround	19.75	**13.75**	14.75	**SRTF**
Avg Response	**2.00**	4.50	8.00	**Round Robin**

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5. Final Conclusion

- **Efficiency:** **SRTF** proved to be the most efficient algorithm for minimizing the overall system delay (Lowest Avg WT and Avg TAT). It is ideal for batch systems where burst times are known.
- **Fairness & Interactivity:** **Round Robin** is the superior choice for interactive systems. While it has higher waiting times, it provides the fastest response (Avg RT = 2.00), ensuring no process is starved.
- **Observations:** * **SRTF** outperforms **SJF** because it can preempt long processes immediately when a shorter one arrives.
    - The choice of **Quantum = 2** in RR significantly improves interactivity but increases context switching overhead.
- **Recommendation:** Use **SRTF** for throughput-oriented workloads and **RR** for user-facing environments like modern operating system schedulers.

Question 2

1. Round Robin (RR) Analysis (Q = 2)

Calculated based on Time Quantum of 2 units.

Metrics Table

Process	AT	BT	CT	TAT	WT	RT
P1	0	3	5	5	2	0
P2	2	11	25	23	12	0
P3	3	5	18	15	10	2
P4	5	6	22	17	11	4
Average				**15**	**8.75**	**1.5**

Ready Queue Sequence

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[P1] -> [P1, P2] -> [P2, P3, P1] -> [P3, P1, P2, P4] -> [P1, P2, P4, P3] -> ...
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2. SRTF (Preemptive SJF) Analysis

Decisions are made every time unit based on the shortest remaining burst time.

Metrics Table

Process	AT	BT	CT	TAT	WT	RT
P1	0	3	3	3	0	0
P2	2	11	25	23	12	12
P3	3	5	8	5	0	0
P4	5	6	14	9	3	3
Average				**10**	**3.75**	**3.75**

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3. SJF (Non-Preemptive) Analysis

Once a process starts, it runs until completion.

Metrics Table

Process	AT	BT	CT	TAT	WT	RT
P1	0	3	3	3	0	0
P2	2	11	25	23	12	12
P3	3	5	8	5	0	0
P4	5	6	14	9	3	3
Average				**10**	**3.75**	**3.75**

(Note: In this specific workload, SJF and SRTF yielded identical results because shorter jobs arrived exactly when the CPU became free).

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4. Comparison Summary

Metric	Round Robin (Q=2)	SRTF	SJF (Non-Preemptive)	Best Performer
Avg Waiting Time	8.75	**3.75**	**3.75**	**SRTF / SJF**
Avg Turnaround	15	**10**	**10**	**SRTF / SJF**
Avg Response	1.50	**3.75**	**3.75**	**RR**

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5. Final Conclusion

1. **Efficiency: SRTF/SJF** provided the most optimal performance for this workload, significantly reducing waiting times compared to Round Robin.

2. **Fairness: Round Robin** maintained its role in preventing starvation, although it introduced more overhead (higher TAT and WT) due to the small quantum relative to process bursts.
3. **Response Observation:** In this specific scenario, RR actually provided better response times because the arrival of short jobs (P3, P4) aligned perfectly with the completion of earlier tasks.

Question 3

1. Round Robin (RR) Metrics (Q = 2)

Processes are rotated every 2 units. This causes frequent context switching.

PID	AT	BT	CT	TAT	WT	RT
4	0	4	6	6	2	0
6	2	6	14	12	6	0
12	5	12	28	23	11	3
9	7	9	22	15	9	3
Average				**14**	**7**	**1.5**

RR Ready Queue Flow: `[P4] -> [P4, P6] -> [P6, P12, P4] -> [P12, P4, P9, P6] -> [P4, P9, P6, P12] -> ...`

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2. SRTF (Preemptive SJF) Metrics

The CPU always executes the process with the shortest remaining burst time.

PID	AT	BT	CT	TAT	WT	RT
4	0	4	4	4	0	0
6	2	6	10	8	2	2
12	5	12	28	23	11	11
9	7	9	16	9	3	3
Average				**11**	**4**	**4**

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3. SJF (Non-Preemptive) Metrics

Processes run to completion once they start.

PID	AT	BT	CT	TAT	WT	RT
4	0	4	4	4	0	0
6	2	6	10	8	2	2
12	5	12	28	23	11	11
9	7	9	16	9	3	3
Average				**11**	**4**	**4**

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4. Comparison Summary

Metric	Round Robin (Q=2)	SRTF	SJF	Best Performer
Avg Waiting Time	7	**4**	**4**	SRTF / SJF
Avg Turnaround	14	**11**	**11**	SRTF / SJF
Avg Response	**1.5**	4	4	Round Robin

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5. Final Conclusion

1. **Efficiency: SRTF** and **SJF** are the winners for throughput, providing the lowest Average Waiting Time (4).
2. **Fairness: Round Robin** is the winner for interactivity, providing the fastest initial response to the user (Avg RT = 1.5), preventing long processes from blocking the system.
3. **Logic Note:** In this specific workload, SRTF and SJF share the same TAT and WT because the arrival times didn't trigger a preemption that would change the final completion order.